Aryo Consulting Group |

Portfolio of Relevant Work

ARYO
CONSULTING GROUP

Project (1 of 8): Redacted full study, ACG built GTM strategy for a 50 year-old insurance company – ranking 10 different lines for them to best enter the Houston market across diff. parameters

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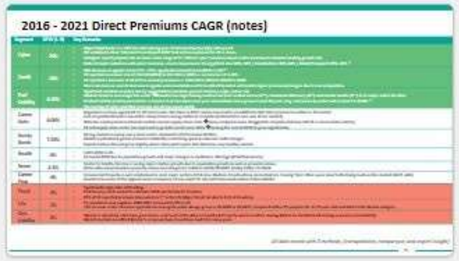

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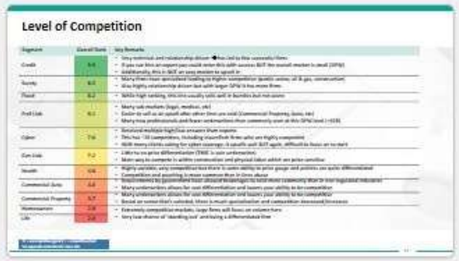

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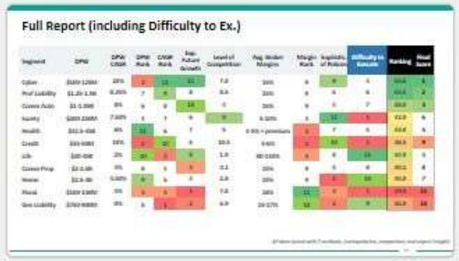

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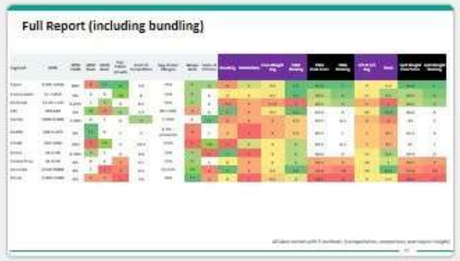

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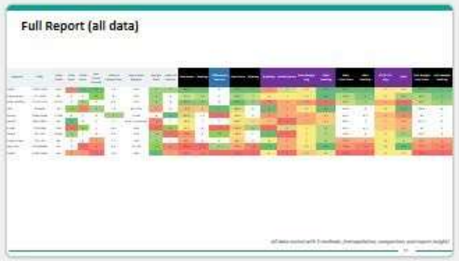

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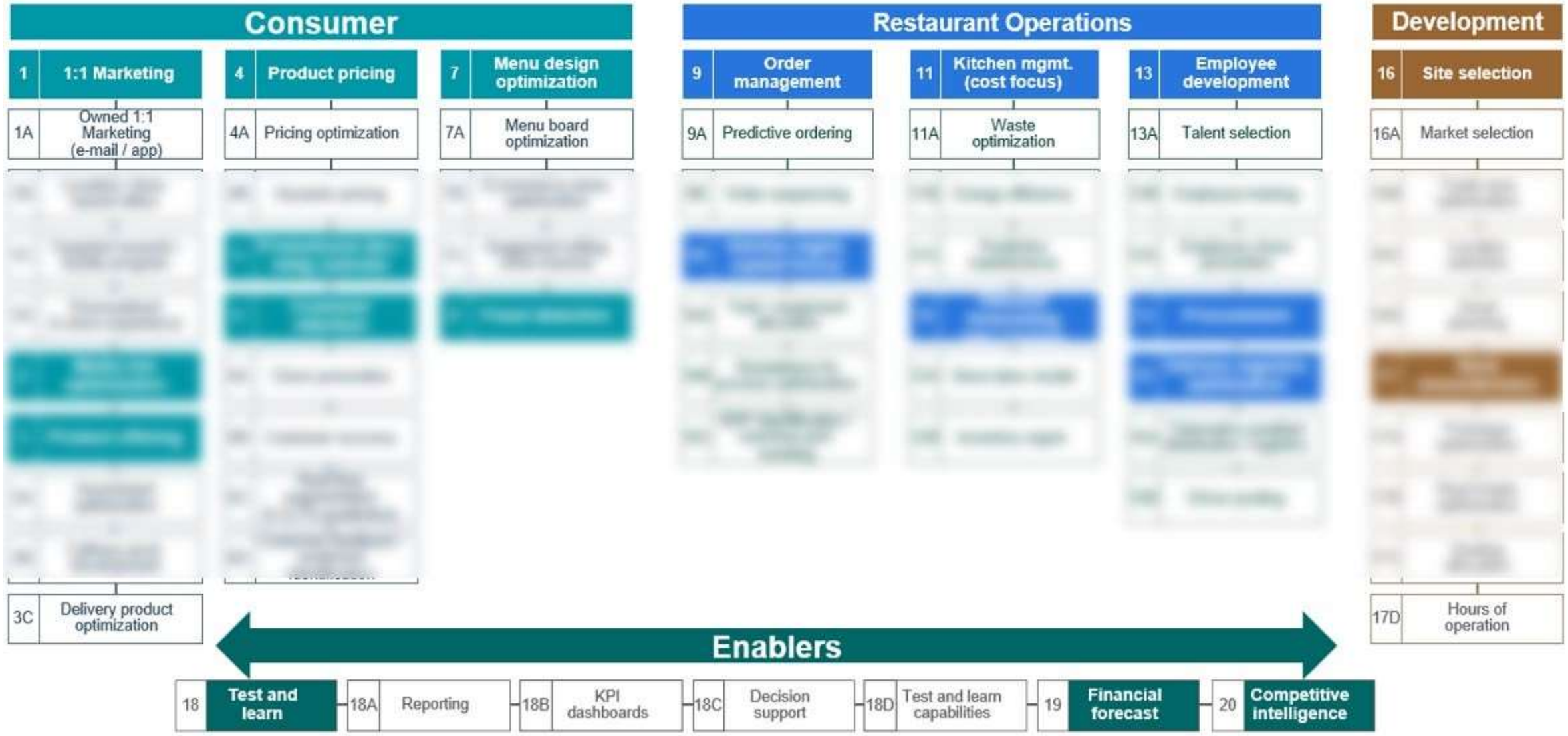
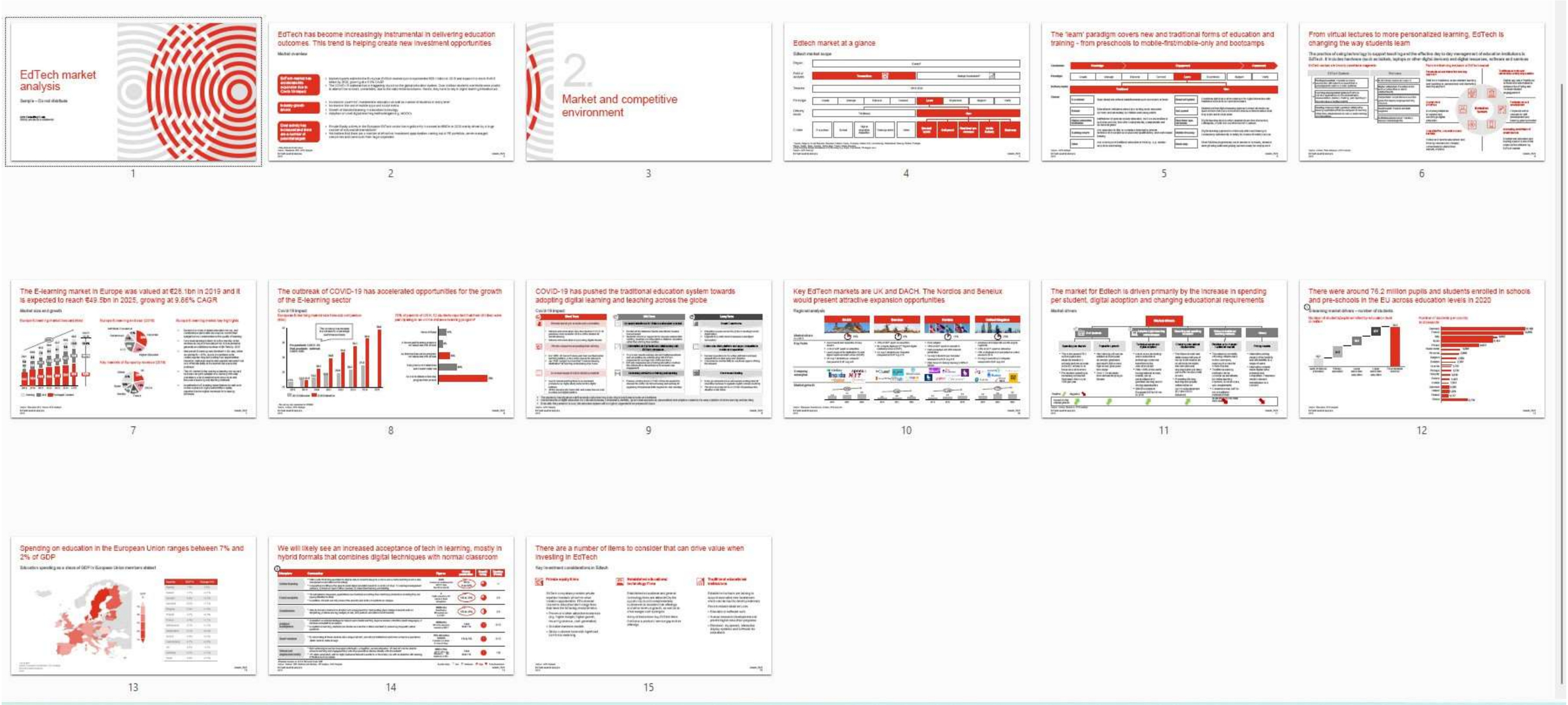
Project (2 of 8): Working with a major VC firm to provide their entire end-to-end strategy of multiple industry's platform's friction point…



Project (3 of 8): Which led to marketing mapping the competitive landscape of various segments the firm was interested in entering

Project (4 of 8): In this project, we analyzed the finances and ran due diligence for a $2B purchase of major EU EdTech company



Project (5 of 8): Worked directly with a leading SaaS and E-commerce company ($50M-100M rev) to identify the financial value of AWS over on-premise cloud software

Cloud-economics: A First and Second Order Financial Perspective on Transitioning from Physical Data Centers to Amazon Web Services (AWS)

Justin Abrams [A1], Jonathan Xu [A2]

Introduction

The data center industry provides infrastructure used in the processing, networking, and storage of data. As the backbone of modern software technology, data centers serve a critical role in expanding capabilities for enterprises. The industry has undergone significant transformation in the past few decades, fueled by relentless growth in technology and digital connectivity. This evolution has facilitated the rise of cloud computing, enabling more efficient and cost-effective data center services. At the forefront of this revolution are large cloud solution providers such as Amazon Web Services (AWS). Through the provision of fast, flexible, and cost-effective cloud-based solutions, these companies have not only reshaped the industry landscape and business model, but have also transformed the way organizations conduct their business operations.

Data Center Industry

Evolution of the data center business model

Mainframes and "traditional" data centers

The data center industry originated in the early

The first commercial data center infrastructure providers emerged, providing organizations with the necessary infrastructure to manage their growing information technology (IT) requirements, while allowing them to avoid the high capital costs associated with building and maintaining data centers in-house.

Colocation and managed service data centers

In the late 1990s and early 2000s, the adoption of the internet and subsequent "dot-com" boom drastically accelerated the need for data centers infrastructure. The industry began to experience significant changes, driven by technological advancements and market forces. Companies began to turn to outsourced data center infrastructure and service providers, leading to the proliferation of several innovative business models:

1. Colocation data centers – Third-party providers that rent or lease space within specialized facilities to host data centers for other companies. In this model, the client company owns the physical infrastructure but avoids the significant capital requirements of building and maintaining a data center. Key players include Equinix and Cyxtera, differentiated in scale of network and geographic coverage.
2. Managed service data centers – Third-

easily launch lightweight, cloud-based products to compete with established incumbents.

- *Operational efficiency* – Cloud computing enables better resource allocation, reduced downtimes, and automation. By outsourcing the IT infrastructure management and support functions, companies can focus on revenue-generating initiatives. Additionally, the resilience of cloud infrastructure mitigates the risk of unplanned downtimes, minimizing lost revenue and productivity.

Unit economic benefits offered by cloud computing varies by business archetype. Specifically, the magnitude of cost reduction is driven by business complexity variables such as company size, data complexity and sensitivity, and IT utilization patterns (e.g., peak usage, amount of data handled) (see Figure 1). For example, a startup can realize significantly larger reductions in overall costs due to the size of IT infrastructure capital requirements relative to other business-related

significant economic advantage by lowering the fixed upfront costs needed to start their businesses.

Enterprise-level companies

For enterprise-level companies, physical on-premise data centers are costly and create significant operational burdens. The total cost of ownership is estimated to be 30-40% higher for traditional data centers compared to cloud solution providers.[5] Cloud adoption not only lowers the high upfront capital investments associated with building and deploying infrastructure, but also reduces ongoing overhead costs associated with operating the physical data centers (e.g., licensing, power, facilities cost). By deploying a cloud solution like AWS, enterprise-level companies can reduce data center infrastructure costs by up to 31% over five years (see Figure 2). Most importantly, cloud computing can drive significant productivity benefits for enterprise-level organizations.

total IT operational costs by up to 51% over a five-year period (see Figure 3).

Small & medium enterprises (SME) and startups

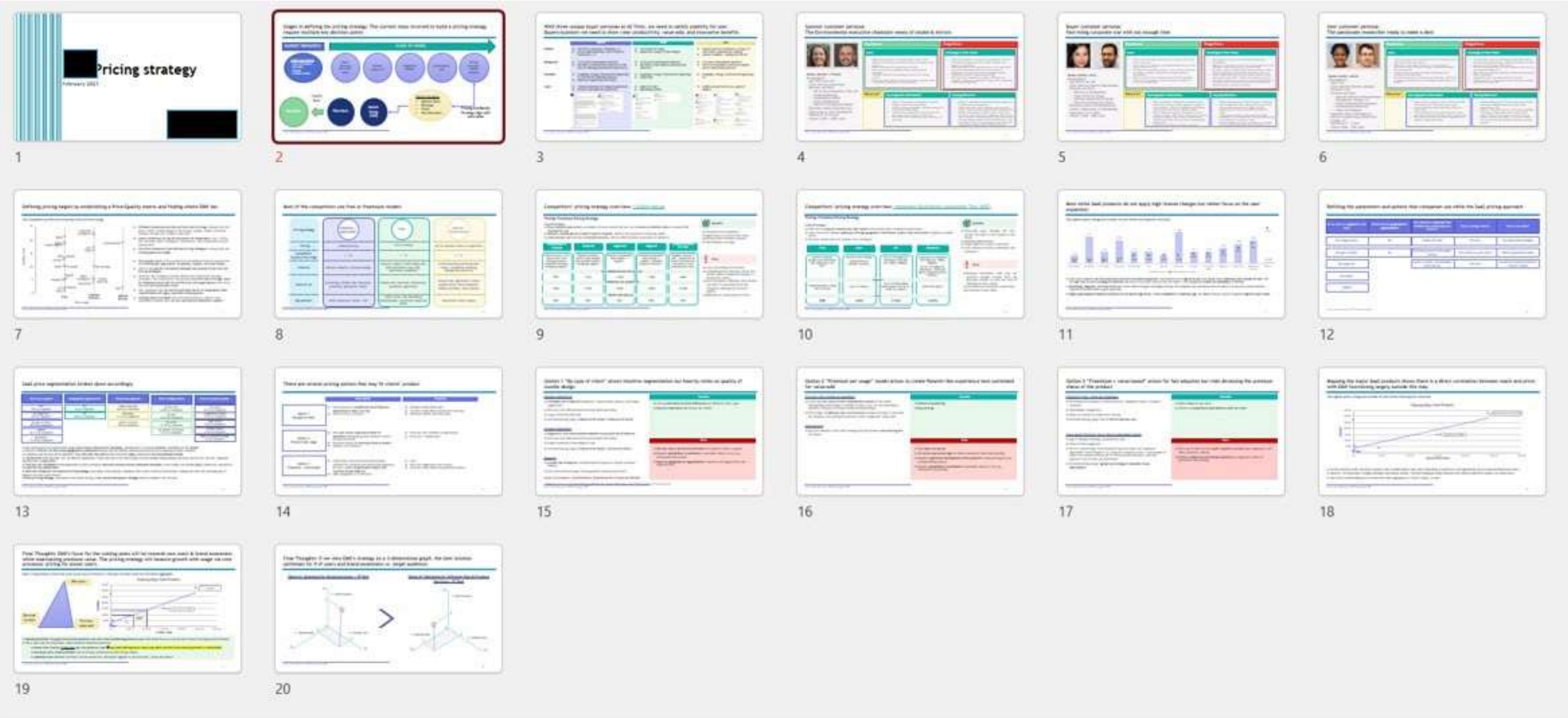
For smaller companies, the cloud provides a significant economic advantage in lowered upfront capital costs. The shift from a capital investment model to an operational expense model can allow startups to realize to up to 90% reduction in data center infrastructure costs.[6] This is further supported in a study conducted by Harvard and Massachusetts Institute of Technology (MIT), which found that AWS lowers the cost of starting new businesses by 15-27%.[7] This greatly reduces barriers to entry, as SMEs and startups are able to launch lightweight, cloud-native products with minimal capital requirements and setup costs. Without high upfront investments in physical infrastructure and IT teams, start-ups can better compete with established incumbents. In a 2019 London School of Economics (LSE) study, 95% of

model allows businesses to scale data center resources up or down based on demand. This ensures that organizations have the capacity and flexibility to handle peak traffic periods without over-provisioning resources during low-demand periods. According to KPMG, the implementation of IaaS cloud solutions can yield 30-60% lower infrastructure costs, driven primarily by drops in "capacity hording".[8]

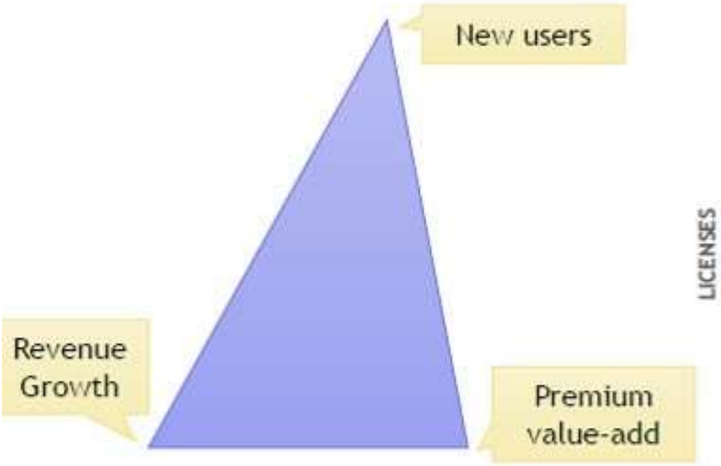
Rapid deployment

Cloud computing dramatically reduces time to market for new products or features, enabling businesses to rapidly deploy applications and services. Through the cloud, companies can quickly develop, test, and launch applications without being constrained by on-premise infrastructure. Companies have indicated up to 75% reduction in time-to-launch for projects implemented through the cloud.[9] This agility allows businesses to respond faster to market trends, customer needs, and competitive pressures.

Project (6 of 8): In this project, we determined the ideal pricing and GTM strategy for a leading Environmental-tech startup and its ability to compare to similar premium SaaS firms

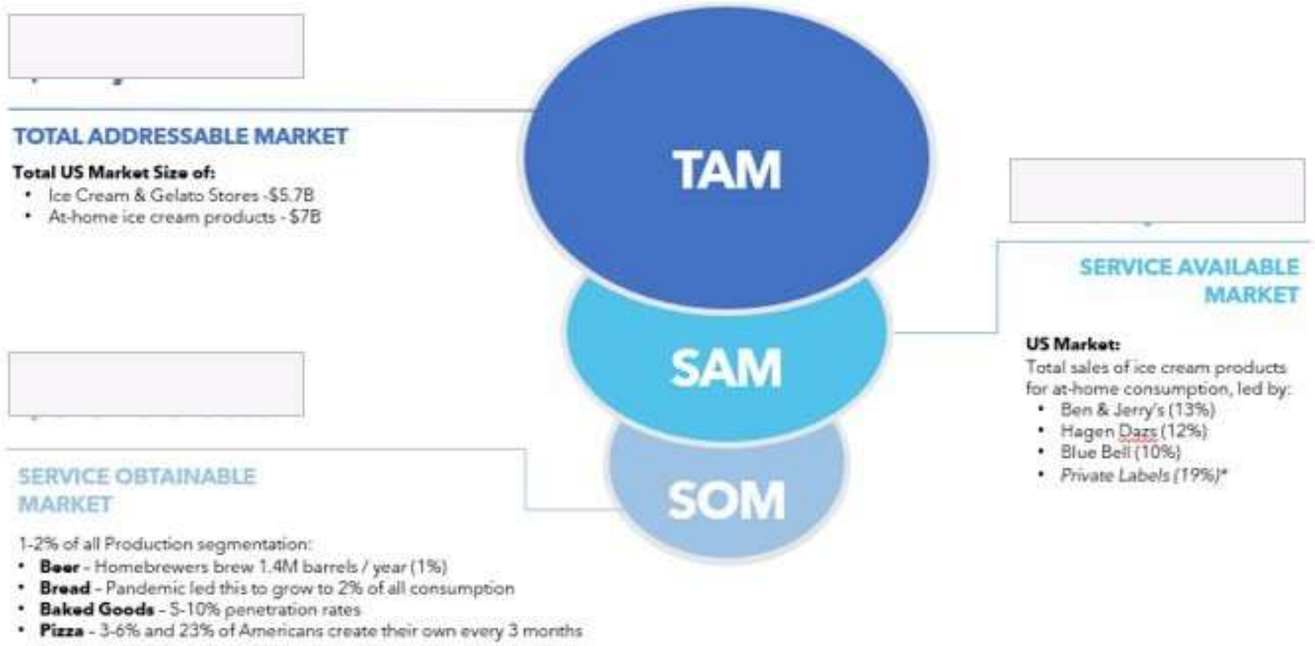


Project (7 of 8): In this project, we determined the ideal pricing and GTM strategy for a leading Environmental-tech startup and its ability to compare to similar premium SaaS firms



Project (8 of 8): In this project, we analyzed a completely new concept on ice-cream to determine the TAM/SAM/SOM along with finding analogs in the current space to define highest penetration rate

			DIY TAM		Penetration %		
Food	**Total DIY $Bn**	**Total Pen Rate**	**Low**	**High**	**Low**	**High**	**TAM ($bn)**
Pizza	$3.0	5%	1.95	3.90	3%	6%	$65
Bread	$0.4	0.75%	0.25	0.50	0.50%	1%	$50
Beer	$1.6	1%	1.25	1.88	1%	1.50%	$125
Cakes	$2.5	8.5%	2.10	3.00	7.0%	10%	$30
Cookies, Brownies, & Other Desserts	$5.5	6%	3.50	4.20	5%	6%	$70



Working with ACG, you're in good hands:
Some of our happy clients include…